



US012408287B2

(12) **United States Patent**
Harvey et al.

(10) **Patent No.:** **US 12,408,287 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **PERCEPTIBLE INDICATORS OF WIRES
BEING ATTACHED CORRECTLY TO
CONTROLLER**

(58) **Field of Classification Search**
CPC .. H04B 3/46; H04B 3/02; H04L 43/50; H04L
67/75; H04L 67/12; H04L 67/125;
(Continued)

(71) Applicant: **PassiveLogic, Inc.**, Holladay, UT (US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/659,869**

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(22) Filed: **May 9, 2024**

(65) **Prior Publication Data**

US 2024/0349437 A1 Oct. 17, 2024

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Related U.S. Application Data

Primary Examiner — James M Perez

(63) Continuation of application No. 17/990,350, filed on
Nov. 18, 2022, now Pat. No. 12,089,360, which is a
(Continued)

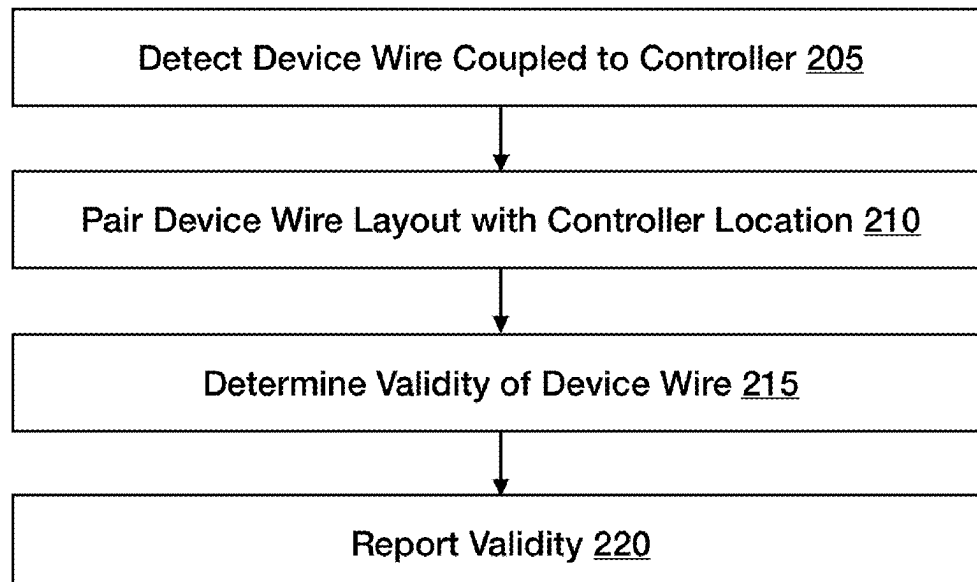
(57) **ABSTRACT**

(51) **Int. Cl.**
H05K 7/14 (2006.01)
F24F 11/32 (2018.01)
(Continued)

A module is described which is slidably attachable to a
controller. Resource wires are connected to the module
through resource connectors. The module has a circuit board
that can perform actions. The module can modify the
function of its resource connectors. These modifications
may be to meet the requirements of resources that are to be
connected to the module. The module may be able to
register, by means of a signal, when a correct resource wire
is attached to the module. The results of such monitoring
may be displayed on a screen associated with the controller.

(52) **U.S. Cl.**
CPC **H05K 7/1465** (2013.01); **F24F 11/32**
(2018.01); **F24F 11/49** (2018.01); **F24F 11/63**
(2018.01);
(Continued)

20 Claims, 9 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/151,365, filed on Jan. 18, 2021, now Pat. No. 11,706,891.

- (60) Provisional application No. 63/070,460, filed on Aug. 26, 2020.

(51) **Int. Cl.**

F24F 11/49 (2018.01)
F24F 11/63 (2018.01)
F24F 11/88 (2018.01)
G01R 31/55 (2020.01)
G05B 13/02 (2006.01)
G05B 15/02 (2006.01)
G05B 19/048 (2006.01)
G05B 23/02 (2006.01)
G06F 1/3209 (2019.01)
G06F 1/3246 (2019.01)
G06F 3/041 (2006.01)
G06F 3/0482 (2013.01)
G06F 3/04847 (2022.01)
G06F 3/147 (2006.01)
G06F 8/41 (2018.01)
G06F 8/51 (2018.01)
G06F 8/53 (2018.01)
G06F 8/74 (2018.01)
G06F 9/4401 (2018.01)
G06F 30/12 (2020.01)
G06F 30/13 (2020.01)
G06F 30/18 (2020.01)
G06Q 30/0283 (2023.01)
H02J 3/00 (2006.01)
H04B 3/46 (2015.01)
H04L 43/50 (2022.01)
H04L 67/12 (2022.01)
H04L 67/125 (2022.01)
H04L 67/75 (2022.01)
H04M 3/30 (2006.01)
H04W 4/80 (2018.01)
H04W 84/00 (2009.01)
G06F 30/392 (2020.01)
G06F 111/04 (2020.01)
G06F 111/16 (2020.01)
G06F 113/04 (2020.01)
G06F 113/16 (2020.01)
G06F 115/12 (2020.01)

(52) **U.S. Cl.**

CPC *F24F 11/88* (2018.01); *G01R 31/55* (2020.01); *G05B 13/0265* (2013.01); *G05B 15/02* (2013.01); *G05B 19/048* (2013.01); *G05B 23/0216* (2013.01); *G05B 23/0264* (2013.01); *G05B 23/0272* (2013.01); *G06F 1/3209* (2013.01); *G06F 1/3246* (2013.01); *G06F 3/04186* (2019.05); *G06F 3/0482* (2013.01); *G06F 3/04847* (2013.01); *G06F 3/147* (2013.01); *G06F 8/436* (2013.01); *G06F 8/51* (2013.01); *G06F 8/53* (2013.01); *G06F 8/74* (2013.01); *G06F 9/4418* (2013.01); *G06F 30/12* (2020.01); *G06F 30/13* (2020.01); *G06F 30/18* (2020.01); *G06Q 30/0283* (2013.01); *H02J 3/00* (2013.01); *H04B 3/46* (2013.01); *H04L 43/50* (2013.01); *H04L 67/12* (2013.01); *H04L 67/125* (2013.01); *H04L 67/75* (2022.05); *H04M 3/305* (2013.01); *H04W 4/80*

(2018.02); *H04W 84/00* (2013.01); *H05K 7/1468* (2013.01); *H05K 7/1477* (2013.01); *H05K 7/1481* (2013.01); *G06F 30/392* (2020.01); *G06F 2111/04* (2020.01); *G06F 2111/16* (2020.01); *G06F 2113/04* (2020.01); *G06F 2113/16* (2020.01); *G06F 2115/12* (2020.01); *H02J 2310/12* (2020.01)

(58) **Field of Classification Search**

CPC H04L 43/00; H05K 7/1465; H05K 7/1468; H05K 7/1477; H05K 7/1481; H05K 7/14; H05K 7/1462; H04W 84/00; H04W 4/80; H04W 4/00; F24F 11/32; F24F 11/63; F24F 11/88; F24F 11/49; G06F 3/04186; G06F 30/18; G06F 30/13; G06F 30/12; G06F 1/3209; G06F 1/3246; G06F 3/0482; G06F 3/04847; G06F 3/147; G06F 30/10; G06F 8/40; G06F 8/41; G06F 8/43; G06F 8/436; G06F 8/53; G06F 8/70; G06F 8/74; G06F 3/01; G06F 3/03; G06F 3/041; G06F 3/0416; G06F 3/0418; G10R 31/55; G05B 15/02; G05B 19/048; G05B 23/0216; G05B 23/0264; G05B 23/0272; G05B 15/00; G05B 19/02; G05B 19/04; H04M 3/22; H04M 3/26; H04M 3/28; H04M 3/30; H04M 3/305; H02J 3/00; G06Q 30/02; G06Q 30/0283; G01R 31/50; G01R 31/55

See application file for complete search history.

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				700/83
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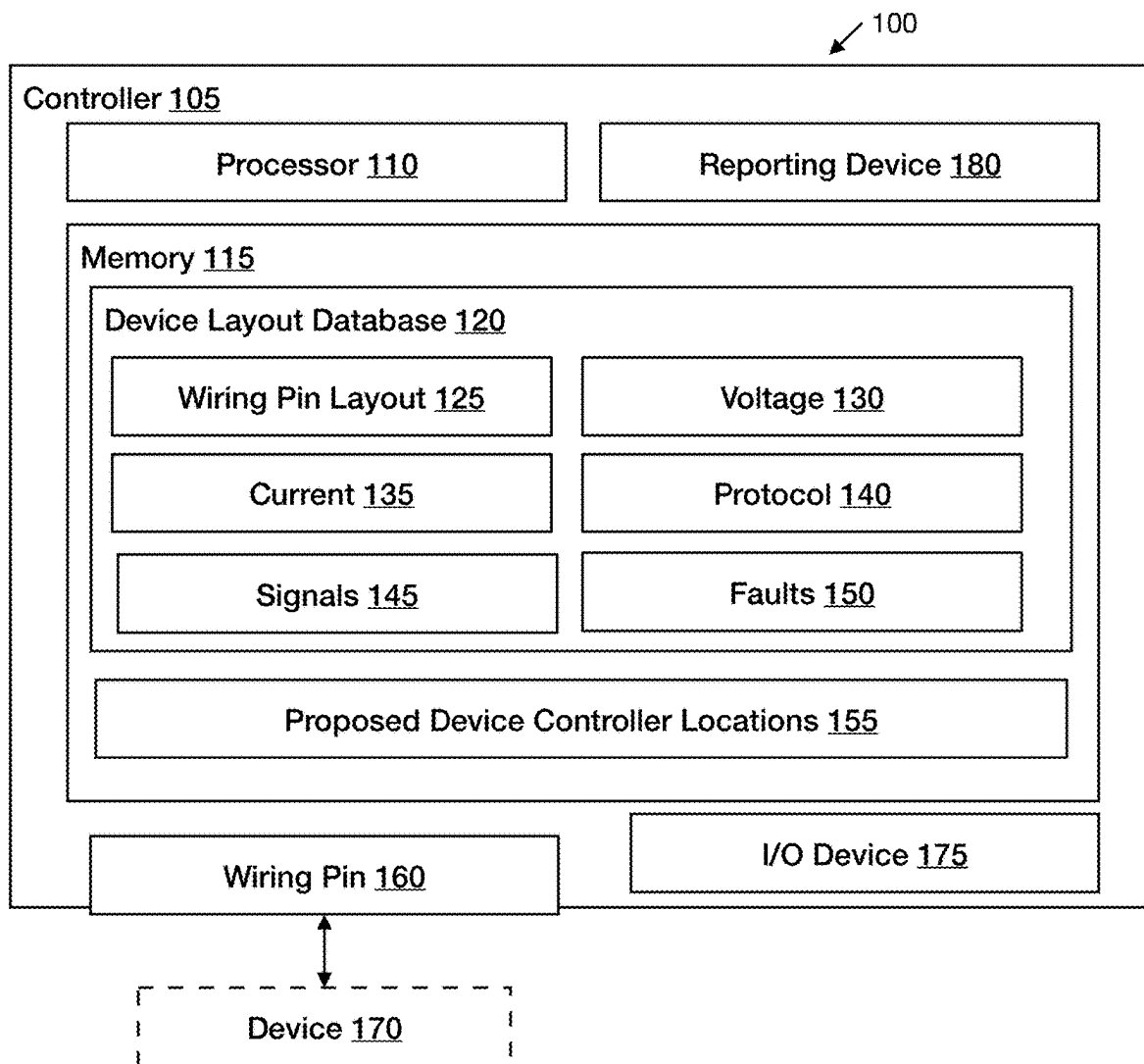


FIG. 1

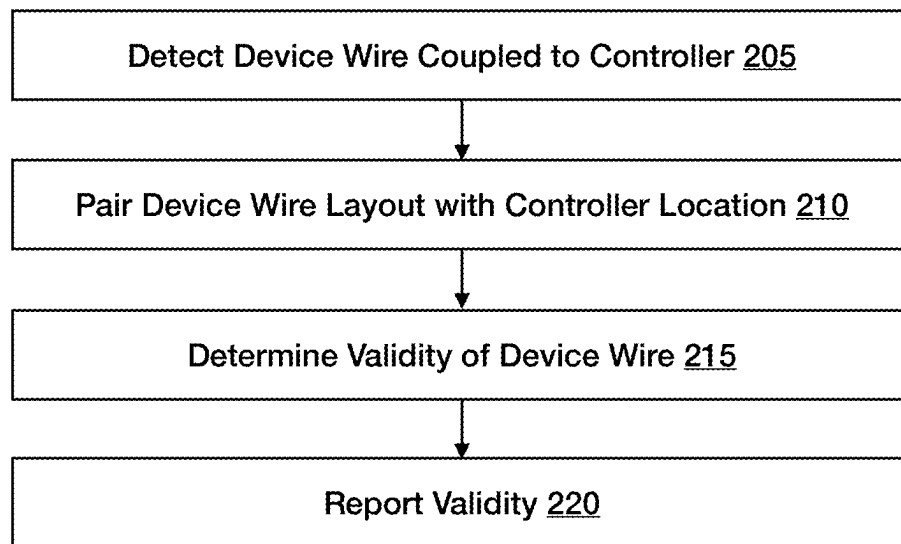


FIG. 2

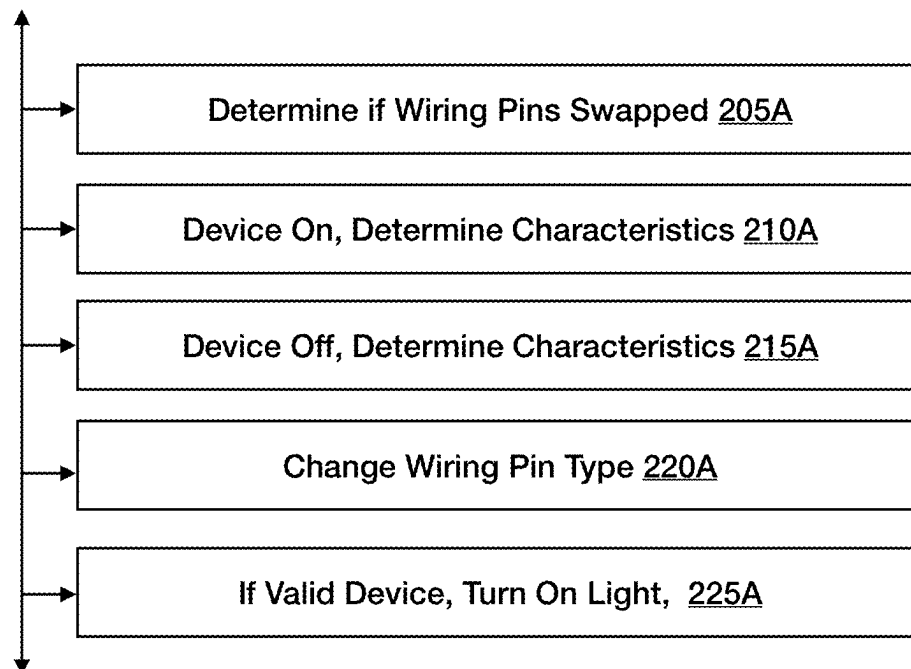


FIG. 2A

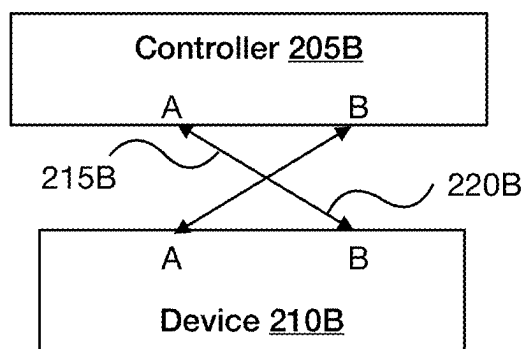


FIG. 2B

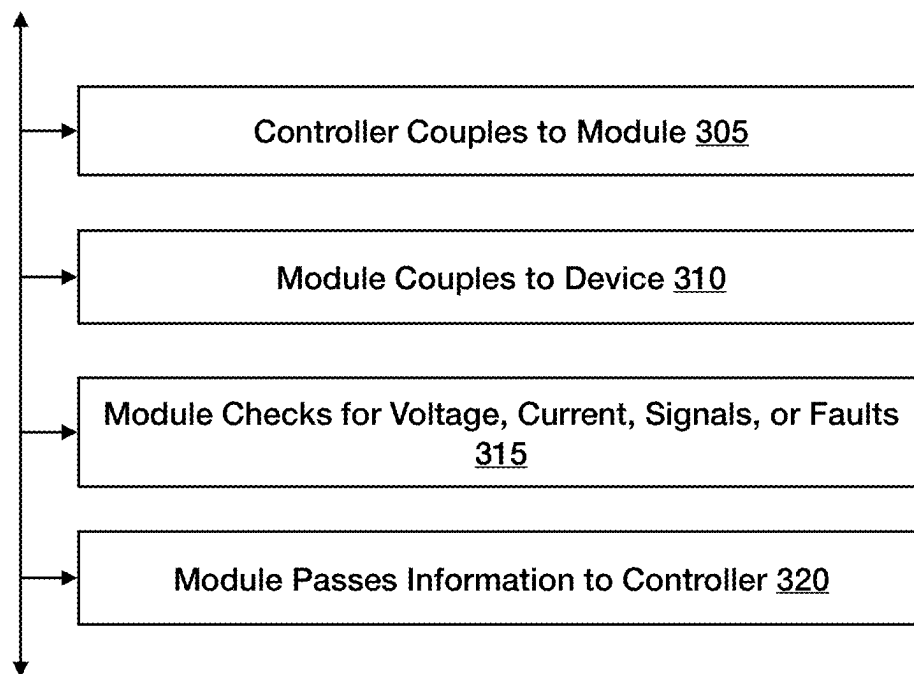


FIG. 3

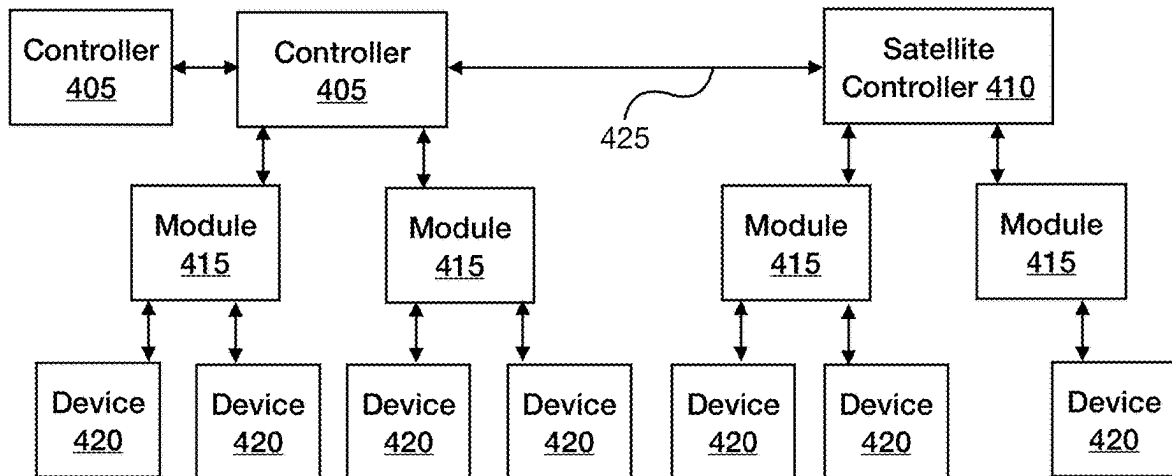


FIG. 4

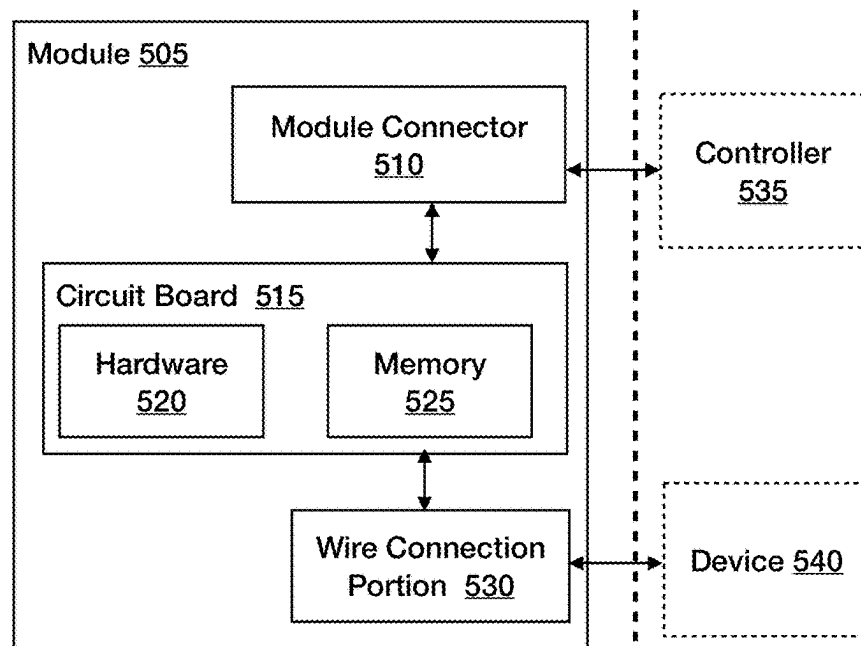


FIG. 5

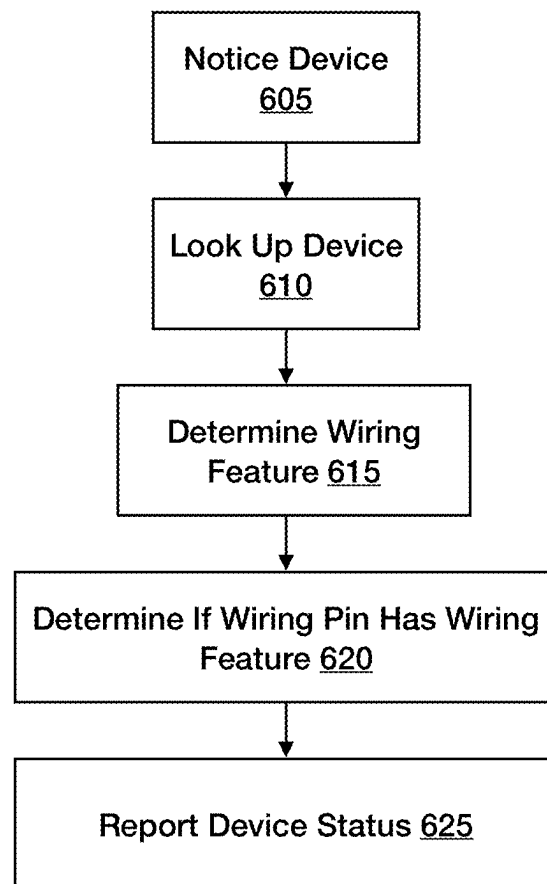


FIG. 6

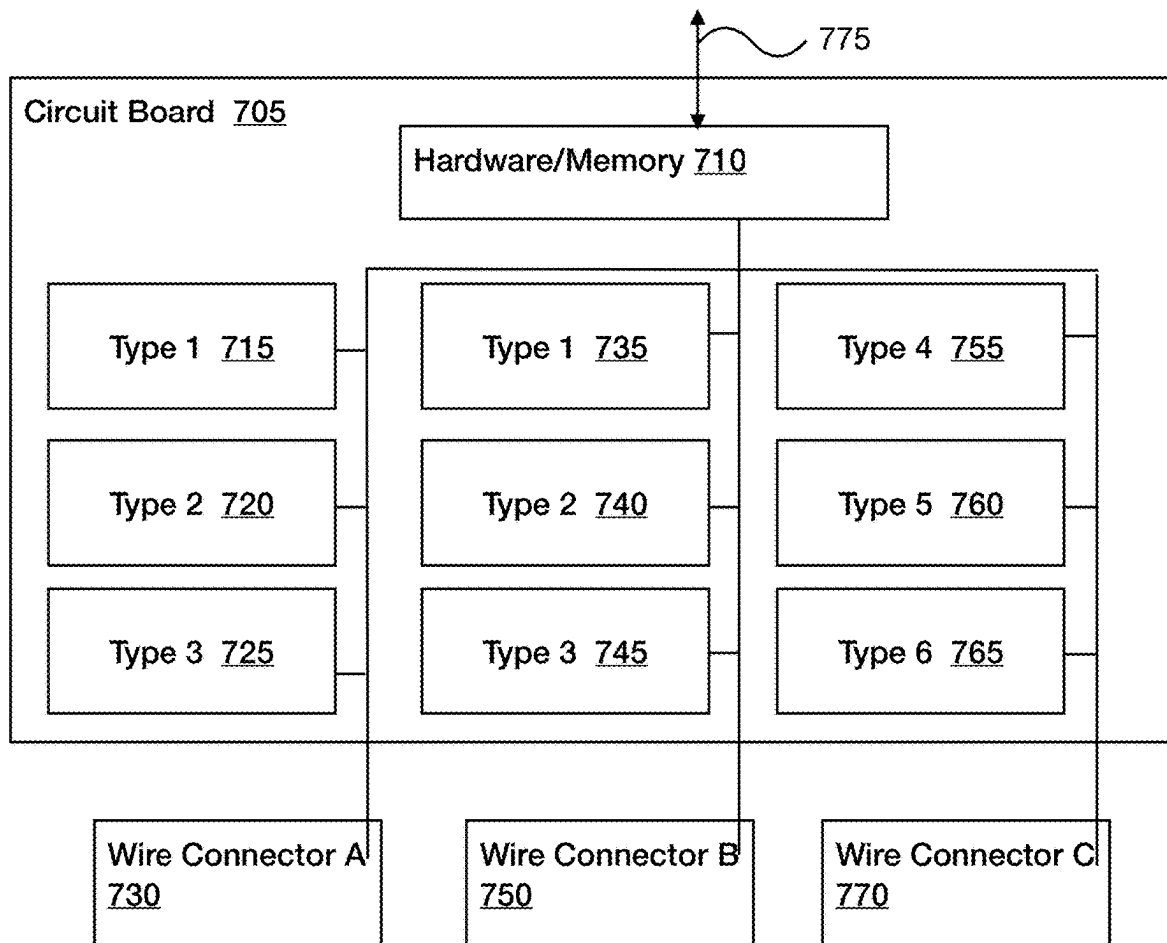


FIG. 7

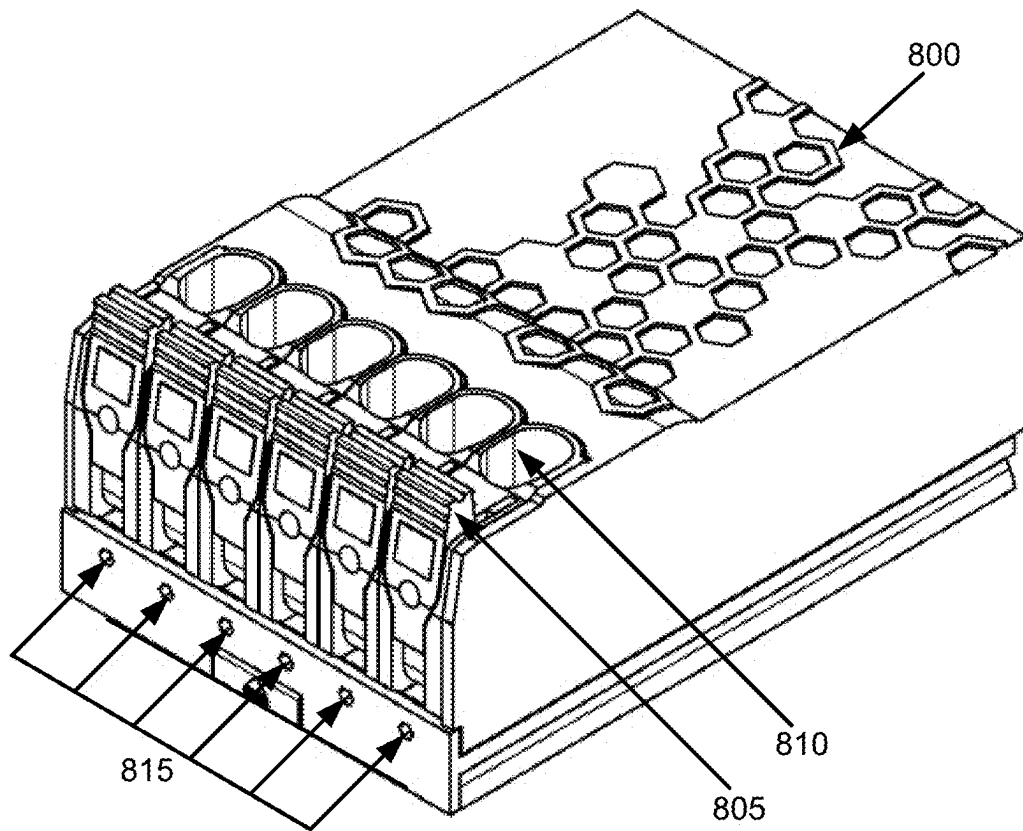


FIG. 8

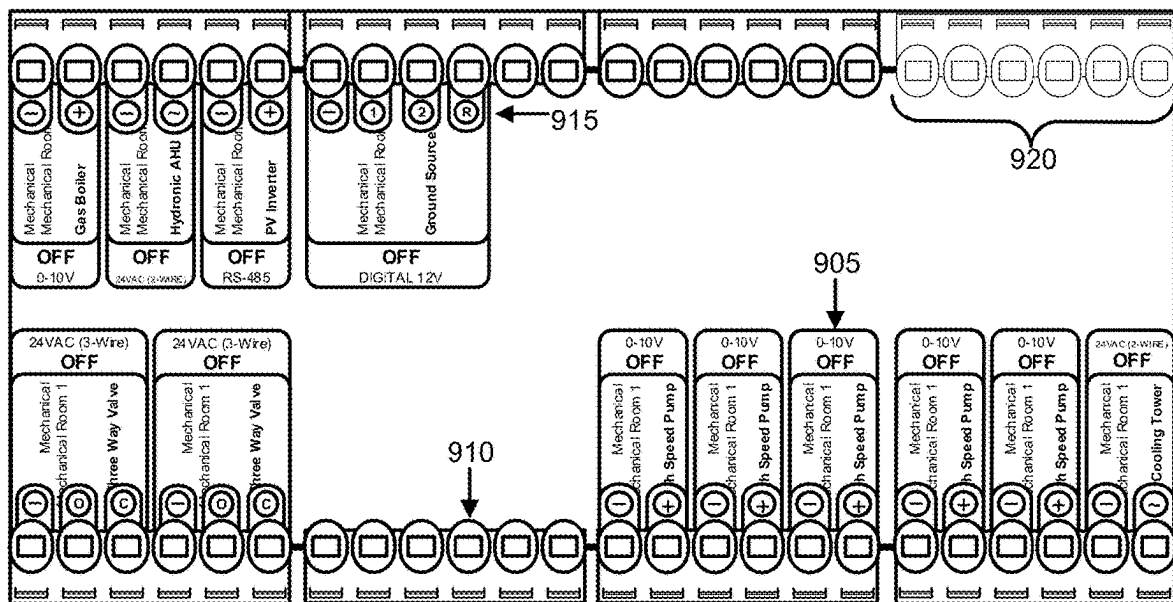


FIG. 9

Text

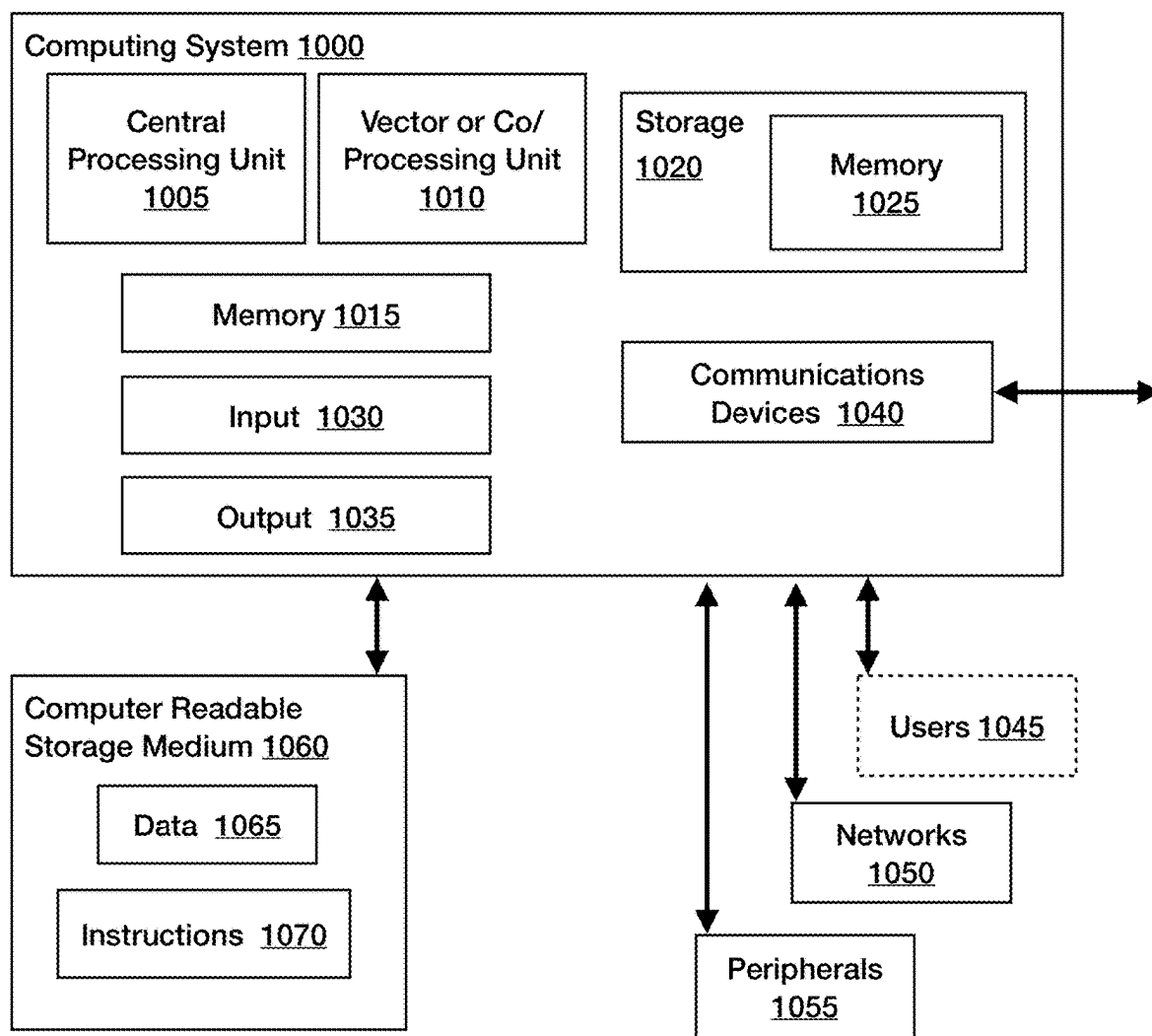


FIG. 10

1

PERCEPTIBLE INDICATORS OF WIRES BEING ATTACHED CORRECTLY TO CONTROLLER

RELATED APPLICATIONS

The present application hereby incorporates by reference the entirety of, and claims priority to, U.S. patent application Ser. No. 17/990,350, filed Nov. 18, 2022, which claims priority of U.S. Pat. No. 11,706,891, filed 18 Jan. 2021, which claims priority to U.S. Provisional Patent Application Ser. No. 63/070,460 filed 26 Aug. 2020.

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FIELD OF INVENTION

The present disclosure relates to line testing device connections to controllers. More specifically, to automatically determining if a correct device line is connected to a controller.

BACKGROUND

Today's "smart buildings" aren't smart at all—just connected. Expensive cloud add-ons that promise to "un-dumb" your control system can only provide low-value results and reduced reliability, all at the cost of more integration effort. Wiring diagrams take hours to create. Further, if a problem comes up when building a structure that requires that a wired device must be moved, the wiring diagram must be recreated from scratch, which can entail quite a bit of effort. When wiring diagrams are changed, the terminals that the devices are to be wired to may also need to be changed as well, which also may entail completely redoing the wiring diagrams from scratch.

SUMMARY

This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description section. This summary does not identify required or essential features of the claimed subject matter. The innovation is defined with claims, and to the extent this Summary conflicts with the claims, the claims should prevail.

In general, some technologies described herein describe a controller system equipped with automated line testing technology. This is used to determine if a device wire attached to a wiring pin on the controller matches the device that the controller expects at that location. The controller comprises a processor, a memory, a database of proposed device-controller locations, a database of device layouts, the device layouts comprising wiring pin layout. To determine if the correct device is attached to a controller, the controller detects that a device wire has been attached to it. The controller determines what device is supposed to be attached at that location, possibly using a database stored in the controller.

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The controller has access to information about the wiring of the proposed device, such as what protocol is supposed to be run, what current is expected to be on the wire at certain times, what voltage is supposed to be on the wire at certain times, what protocol is supposed to be used by the device to send and receive messages from the controller, etc. Using this information, the controller determines what information is being sent on the wire and if that information matches up with what it expects. The validity (if the device wire is the correct device or not) is then reported. In some embodiments, if the device is the one that the controller expects, then the validity is true. If the device wire is not the one that the controller has recorded that is supposed to be attached at that location, then the validity is false.

Some innovative embodiments are directed to automatic line testing which allows a user to insert a wire in a module associated with a controller, and a light on the module associated with the wire terminal insertion location indicates that the correct wire has been inserted. Some embodiment may indicate that an incorrect wire has been inserted. In some implementations, the device may need a different sort of wire terminal insertion location, in which case, in some implementations, the controller may be able to direct the module to change wire terminal insertion types to match the type needed by the device.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a functional block diagram showing an exemplary embodiment of a controller in conjunction with some described embodiments can be implemented.

FIG. 2 is an exemplary flow diagram of illustrating some steps in some implementations of automated point mapping in conjunction with which some described embodiments can be implemented.

FIG. 2A is an exemplary diagram that describes controller-device interaction.

FIG. 2B is an exemplary diagram that further describes controller-device interaction.

FIG. 3 is an exemplary diagram that further describes controller-module interaction.

FIG. 4 is an exemplary flow chart illustrating steps in some methods of automated point mapping generation with which some described embodiments may be implemented.

FIG. 5 describes some more details on some module implementations.

FIG. 6 is an exemplary flow diagram illustrating steps in some methods of automated point mapping with which some described embodiments may be implemented.

FIG. 7 is an exemplary diagram that describes module details with which some described embodiments may be implemented.

FIG. 8 is an exemplary diagram that further describes module details with which some described embodiments may be implemented.

FIG. 9 is a block diagram of an exemplary computing environment in conjunction with which described embodiments can be implemented.

FIG. 10 shows some components that may be included in a computing environment with which described embodiments can be implemented.

DETAILED DESCRIPTION

Disclosed below are representative embodiments of methods, computer-readable media, and systems having particu-

lar applicability to automated line testing. Described embodiments implement one or more of the described technologies.

Various alternatives to the implementations described herein are possible. For example, embodiments described with reference to flowchart diagrams can be altered, such as, for example, by changing the ordering of stages shown in the flowcharts, or by repeating or omitting certain stages.

I. Overview

When originally designing and building a structure, the creation process can include designing the structure using a portion of the controller system that will eventually control the various devices (such as HVAC equipment, sensors, safety systems, etc.). In such a system, the specific wiring details and other characteristics about the devices will be input into the controller system during the design process. The wiring diagram for the building can then be created by the controller system. This wiring diagram can comprise the specific wiring diagram for each controller in the building. The specific wiring diagram can include the wiring pin(s) in a controller that a device will be attached to. When a device and its details are input into the original design, the controller already then knows enough about the device to be able to run it properly. Thus, the controller automatically knows what devices were wired to which locations, and what their specifications are, down to their wiring requirements. Thus, when a device is physically being wired to a controller during the building process, the controller can tell if the correct wire of the device is being attached to the correct spot on the controller.

The technical character of embodiments described herein will be apparent to one of ordinary skill in the art, and will also be apparent in several ways to a wide range of attentive readers. Some embodiments address technical activities that are rooted in computing technology, such as more quickly and efficiently constructing electrical, HVAC, etc. systems, etc. Further, because the controller system itself knows the devices that are attached and their protocols, this information is automatically historically archived. Other advantages based on the technical characteristics of the teachings will also be apparent to one of skill from the description provided.

FIG. 1 illustrates aspects of a system architecture which is suitable for use with some automatic line testing embodiments. A system 100 to allow automated point mapping line testing is disclosed. The system 100 includes a controller 105 which comprises computer processor 110, memory 115, and a wiring pin 160 that can be used to automatically line test devices 170 wired to the controller 105. The memory 115 can be any appropriate volatile or non-volatile storage subsystem. For example, the external memory can be volatile memory, e.g., static memory cells, as in FPGAs and some CPLDs; or non-volatile memory, e.g., FLASH memory, as in some CPLDs, or in any other appropriate type of memory cell. The wiring pin 160 comprises a connector between the controller 105 and a device, such that the device can be wired to the controller and the controller can then pass messages to the device and receive messages from the device. Examples of such messages are “turn on,” “turn off,” “turn on low,” “determine how much power is being drawn,” and so on. The wiring pin is any sort of connector that can pass information to a device, from a device, or to and from a device. For example, the wiring pin may be a terminal

contact such as a pogo connector. The wiring pin 160 may be a part of a module that is itself connected to the controller 105.

The memory 115 has stored in it a device layout database 120 which itself comprises a wiring pin layout 125 (e.g., describing which pins are expected to be attached to which devices, and which pin of each device attaches where). The controller, through the device layout database and other associated databases, also understands the nature of each of the devices that are to be attached to it—the protocol—such that it also has stored in memory at least one of a current 135 expected for a pin that is to be attached to a device, a voltage 130, the protocol 140 that the device is running, e.g., a standard set of signals that the controller can expect to receive from the device, and what signals the device expects back. Some devices may have an idiosyncratic signal set, or a single signal 145 that may also be stored. Some controllers or devices also have a database of faults 150 that may be associated with devices or with other controller features, such as network connections. Even though items 125-150 are listed in the device layout database, this is for convenience only; they can be stored in separate databases, etc. Protocol should be understood as a broad term encompassing open protocols such as BACnet, closed protocols, such as SNA, protocols for individual pieces of equipment, such as whether they are analog, the voltage allowances, e.g., 0-10 volts, a single value such as 40 milliamps or 12 volts; the signals that a specific piece of equipment uses to communicate with a controller, such as on/off values, current, voltage, networkable protocol, modulation, frequency, any combination of the above, etc. The controller memory

The device layout database can take many forms, as understood by those of skill in the art. With reference to FIG. 4, an example device layout is shown. In a database, (such as one that may be stored in memory 115) controllers (such as that shown at 410) has connections to modules 415 that are at specific locations within the controller. Each of these modules has connections to devices 420 that are at specific locations within the modules. The devices may have wiring information stored about them in memory associated with the controller; everything may have location within a structure stored with it, also. These devices may be any sort of device that can be wired to a controller, such as HVAC devices, sensors, entertainment systems, security systems, irrigation systems, and so forth. In some implementations, with multiple controllers, a master controller may store all the device layout database information 120; in other implementations the device layout database 120 may be distributed between multiple controllers. In some implementations, a satellite controller 410 can fit modules associated with devices, but the information about the satellite controller devices is stored in memory 115 associated with a controller leader 410.

As with the device layout database, the controllers and modules may also be laid out in a manner such as shown in FIG. 4. Multiple controllers 405, 410 may be connected 425 through wired connections, wireless connections, or a combination of both. In some systems, a master controller (not shown) divides the work in the network between the controllers in the system using distributed system methods. In some systems, the satellite controllers are not given any work except that that concerns their own devices. In some systems, controllers 405 can be connected to one or more modules 415, which are themselves connected to devices 420. The devices 420 then connect to the controllers 405 through the modules 415.

Also stored in memory may be proposed device controller locations **155**. Each controller has a certain number of wiring pins **160** that have devices **170** assigned to them. These assignments are accessible to the controller such as in e.g., a device layout database. The controller-device layouts may be input by a user using an I/O device **175** associated with the controller in any manner of ways. For example, when initially designing a building, a user may digitally input the desired devices into a digital version of the building, along with the devices wiring information. In some embodiments, common devices may have wiring and other electrical information already input into a database about them, so a user may just need to specify the manufacturer and specific type of the device. For devices not in the database, users may be able to input the necessary wiring information themselves. Once the digital devices are input into the digital building, a controller, a controller system, or another sort of computer system associated with the controller may then create the controller-device wiring layout, including the wiring locations for the devices within controllers. Users may be able to change these associated wiring layouts as needed. In some implementations, users may input how the devices fit into the controllers themselves. In some implementations, users may input some device locations within the controller system, while letting the controller system design other controller-device layouts. In some implementations, users may directly input the devices that are to be attached to controller wiring pins. Other methods known by those of skill in the art may be used, as well.

A reporting device **180** is also disclosed. In some embodiments, the reporting device reports on whether the wiring pin **160** has been attached to the correct device **170**. This reporting device **180** may be any sort of indication that an event has occurred.

With reference to FIG. 2, a flowchart is shown that may be used in some systems and methods. Steps in an embodiment may also be done in a different order than the top-to-bottom order that is laid out in FIG. 2. Steps may be performed in parallel, serially, or in a partially overlapping manner. Steps may also be combined, omitted, renamed, regrouped, or otherwise depart from the illustrated flows, provided that the process performed is operable and conforms to at least one claim.

At **205**, the controller detects that a device wire has been coupled to the controller at a controller location. The controller may be a single controller, or a distributed system of controllers, etc. These controller locations may be a wiring pin **160**. The controller location may also be a wiring pin associated with a module **415**. The controller location may be some other location where a device can be connected, possibly through an intermediary, to a controller. Because the controller (through, e.g., the device layout database) understands which device is connected to its wiring pins, it can determine which device will be expected to be attached to a given wiring pin. These wiring pins may be directly attached to a controller or may be attached to a module which is attached to the controller. (For more on modules, see FIGS. 3, 4, and 5 and the accompanying text.)

At **210**, the controller determines which device, and possibly, which wire of which device (for a multi-wire device) should be attached to that wire. This may be done by pairing a device layout in the database with the location on the controller in the database that the unknown device has plugged into. At **215**, the validity of the device wire is determined. The device layout database comprises information on how the device works, including wiring information. The controller can check the wire for such information as is

necessary to determine if the correct device is at the other end of the wire-if the device is valid. Determining validity of the device may entail checking that the correct current is on the wire, the correct voltage is on the wire, the wire is following the expected protocol, a signal received on the device wire, a signal received on the device wire in response to a signal sent by the controller, etc. Other ways that may be used to determine if a device wire is valid includes using the device wire to turn the device on; checking if the correct voltage is running through the wire when the device is on; checking if the correct current is running through the wire when the device is on; checking if the correct power (current*voltage), etc. a device is detected a device layout is paired to a controller wiring pin. Because the controller (through, e.g., the device layout database) understands which device is connected to its wiring pins, it can determine which device will be expected to be attached to a given wiring pin. These wiring pins may be directly attached to a controller or may be attached to a module which is attached to the controller. (For more on modules, see FIGS. 3, 4, and 5 and the accompanying text.)

In some embodiments, the device layout database comprises a protocol associated with the device **170**. This may be a protocol is used by the controller to speak to the device, and may be the protocol that the device uses to speak to the controller. These may be the same or different protocols. These protocols may be protocols that are well understood, or may be custom protocols that are input into the controller, specific for a given device.

At **220**, the validity of the device wire is reported. This report may be made in a dedicated reporting device **180**. The reporting device may have a signal that indicates that the correct device has been attached to a wiring pin. The reporting device may have a signal that indicates that an incorrect device has been attached to a wiring pin. The reporting device may have a series of lights. The reporting device may be text that is displayed on an I/O device. The reporting device may be an indication that is displayed on an I/O device. The reporting device may be text that is displayed on an I/O device. A certain color light at the reporting device may signify that the correct device has been wired to the wiring pin. A different color may indicate that an incorrect device has been wired to the wiring pin. The reporting device may be a light associated with a wiring pin that lights up green when a correct device **170** is wired to a wiring pin **160**. The reporting device may be light associated with a wiring in that lights up red when an incorrect device is wired to the wiring pin **160**.

With reference to FIG. 2A, at **205A**, in some embodiments, it is determined if connector pins are swapped on a device or among devices associated with a single module or among devices associated with a single controller. Because the controller understands the expected inputs from each of the connectors that are expected to be connected to it, if connector A is exhibiting the protocol of connector B, and vice-versa, the controller can determine that connector A and connector B have been swapped.

At **210A**, in some embodiments, the controller turns a device on, using information from an information source, such as a device layout database. Once the device is on, the controller can send information along the device wire allowing the controller to determine characteristics of the device. These characteristics may be device-dependent, but may be used to determine if the correct device is coupled to the controller. At **215A**, in some embodiments, the controller determines a device is off, using information from an information source, such as a device layout database. Once

the device is determined to be off, the controller can check voltage, current, etc., of the device wire allowing the controller to determine characteristics of the device. These characteristics may be device-dependent, but may be used to determine if the correct device is coupled to the controller.

At 220A, in some embodiments, the controller changes wiring pin type 220A. Sometimes the controller and the type of the device connection (e.g., a wiring pin) do not match. In some embodiments, the wiring pin features are encoded in hardware and software, and/or encoded in software that can make hardware modifications such that the controller can change the pin to match that required by the device by modifying software associated with the pin. In some embodiments, the pin is associated with a module that itself is associated with the controller. In some embodiments, the controller gives software instructions to the module associated with the pin, which in turn, modifies its hardware to change the pin to the appropriate type. Encoding such sorts of hardware features in software is well-understood by those of skill in the art. Basically, a series of hardware features exist, and the software turns the needed ones on, and the ones that are not needed off, such that the wiring pin has the desired set of features. The type of protocol that the pin understands can also be changed in the database files associated the pin. In some embodiments, the controller has modules that the devices are wired to. These modules have circuit boards that can have specific features associated with the pins on the module turned on and off at the direction of the controller.

At 225A, if the device has been determined to be valid, a light associated with the wiring pin is turned on. In some embodiments, this light is green. In some embodiments, the light is a different color.

FIG. 2B shows an example of a device 210B with wiring pins swapped on a controller 205B. Device 210B is wired to a controller 205 with multiple wires 215B, 220B. As the controller understands the signals and current/voltage that are expected to be received on the wires, in some embodiments the controller can determine when the wiring has been swapped, that is, as shown in this example, device wire A was expected to be wired to controller wire A but instead has been wired to controller wire pin B, and vice versa. This also works for devices with more wires. In some embodiments, the controller can determine when a device A at wire A has instead been wired to controller location C. In some embodiments, when a device A is expected to be wired to locations A, B, and C, but is wired elsewhere within the controller, the controller will detect this and determine the correct location for the device.

With reference to FIG. 3, in some implementations the controller is operationally able to couple to a module 305, and the module is operationally able to couple to the device connector 310. In some implementations, the module then checks for at least one of voltage, current, signals, or faults 315. In some implementations, the module then passes the learned information (voltage, current, faults, signal) to the controller 320. The module provides a connection between the device connector and the controller, such that information about the device (e.g., voltage, current, faults, etc.). In instances where the specific details about a connector/wiring pin needs to be changed (such as discussed above with reference to "a partial list of possible wiring pin features"), the hardware that controls pin behavior may reside on the module. The controller, in such instances, will determine which changes need to be made to the pin, and then pass information on to the module about how the module needs to make changes to itself.

An illustrative example of a controller-module hierarchy can be seen with reference to FIG. 4. At the top of the hierarchy is one or more controllers, such as the controller shown in FIG. 1 at 105. The controllers may be able to communicate 425 with other controllers 410 in a controller system using a wired connection, a wireless connection, or a combination of the two. A satellite controller may be a controller that is lacking some functionality. For example, a regular controller 405 may be able to hold 6 modules 415, while a satellite controller 410 may be able to hold 2 modules. Other possible differences may be that the satellite controller has less memory, 115, different computer hardware 110 such as processors, etc. As already mentioned, the controllers may hold a certain number of modules 415. These modules then connect to devices 420. The device 420 can be thought of connecting to the controllers 405, 410 through the module 415. Different controllers may hold different number of modules; different modules may hold different numbers of devices. Some modules may be able to accommodate six wiring terminals.

FIG. 5 describes some more details on some module implementations. In some implementations, the module 505 further comprises hardware that has the operational ability to determine a voltage, a current, or specific fault detection, which information it then passes to the controller. A controller 535 may send messages to a module connector 510. The module connector may then send those messages to its circuit board 515, which may then process the messages and make decisions using its hardware 520 and memory 525. This may result in an altered signal from the signal originally sent by the controller. This altered signal may then be passed to a device connection portion 530 (e.g., a wiring pin 160) which sends the signal to a device 540. This process may work the opposite way as well. A device 540 sends a message through a device connector (wire connection portion) 530 to the circuit board 515, which may then process the message, changing the signal. The changed signal is then sent through the module connector 510 to the controller 535. In some embodiments, the signal is not changed from the controller 535 to the device 540. In some embodiments, the signal is not changed from the device 540 to the controller. In some embodiments, the circuit board 515 can change the nature of the device connector 530 that is connecting to device 540 depending on the device 540's requirements. In some implementations, the module, using its circuit board, can determine the voltage, current, and/or fault on the device connector (wiring pin) 530 attached to device 540.

A partial list of possible wiring pins 160 features that exist in hardware and can be changed to other features in software are shown below.

- Ground connection (through a FET)
- 24 VAC power connection (through a Triac)
- 0-10V DC analog voltage (via a DAC)
- Strong 1-wire pullup (via a FET)
- High-range voltage divider: don't know what will be plugged in, so anticipate highest tolerated (480 volt) if someone plugs in an unknown signal with unknown voltage measure first with high range ability, sees a really tiny signal, can switch to low range, measure small signal, can be safe the whole time.
- Low range voltage divider
- 4-20 mA current shunt
- 24 VAC current shunt
- DC offset injector that can measure both halves of an AC waveform
- Electromechanical relays, each relay having 3 connections (normally open, common, normally closed)

120 VAC outputs

Chips that measure current

A feature that a can switch 250 volt AC up to 10 amps
Control for high-current motors.

In FIG. 6 a flowchart is shown that may be used in some methods. Steps in an embodiment may also be done in a different order than the top-to-bottom order that is laid out in FIG. 6. Steps may be performed in parallel, serially, or in a partially overlapping manner. Steps may also be combined, omitted, renamed, regrouped, or otherwise depart from the illustrated flows, provided that the process performed is operable and conforms to at least one claim.

At 605, a controller (e.g., 105) notices that a device has been operably attached to a controller wiring pin. This device may have been attached directly to the controller, or may have been attached to a module that itself is attached to a controller (as seen with reference to e.g., FIGS. 4 and 5 and the accompanying text). At 610, the controller consults a device database to determine which device is supposed to be hooked up to that location. If a device has multiple connectors, the controller may look up each pin separately, may look the device up all at once, etc. At 615, the controller determines a wiring feature of the device. The controller may look this feature up in a device database separately, or may have looked it up initially when it was determined which device was to be hooked up to that location. At 620, the controller determines if the controller wiring pin has the feature(s) that were looked up. It determines its validity, that is, if it had the expected feature (or features) or not. This feature may be any sort of feature that can determine if the correct wire has been hooked up, such as if the correct voltage is on the wire, the correct current, if the correct protocol is being used, or a special set of signals may be sent to the device that causes the device to identify itself to the controller if it is the correct device. Other methods of detection may be used as well. If the feature is (or features are) present, or behave as expected, etc., then the device is valid. If the feature(s) are not present, or don't behave as expected, then the device is invalid.

At 625, whether the device passed or failed the line testing is reported using a reporting feature. This feature may be a signal. In some embodiments this might be light close to the wiring line that is being tested. It might turn red if the wrong wire is installed, and turn green if the correct wire is installed. In some embodiments the reporting feature may be a signal on a user interface screen associated with the controller, in some embodiments, the feature may be a report; in some embodiments, the feature may be a sound indicator, etc.

As shown with reference to FIG. 7, in some embodiments, a module can change the wiring pin type to match wiring type of the device. In some embodiments, the module has a circuit board 705 with hardware and memory 710. In the described module, there are three wire connectors, wire connector 1 730, wire connector 2 750, and wire connector 3 770. These wire connectors may be of different functions, e.g., type 1 through 6. A possible list of possible wiring pin connector types is shown elsewhere. The circuit board has hardware associated with the wire connectors that can be enabled by the module receiving the appropriate signal from the associated controller, allowing the wire connectors to be of one or more of those types. Wire connector A 730 has, associated it in the circuit board hardware for 6 types: Type 1 715, Type 2 720, and type 3 725. Wire connector B, similarly has the hardware potential to be one of the three types as well-Type 1 735, Type 2 740, and type 3 745. Wire connector C has different types associated with it: Type 4

755, Type 5 760, and type 6 765. The controller send a signal (or a program or another indication) to the module that wire connector A is expected to be type 3. The module may then be able to use its hardware/memory 710 on its circuit board to send a signal 775 telling a Wire Connector A to be of Type 3. The module may be able to connect wire connector type 3 725 to Wire connector A, making Wire connector A of type 3. In some embodiments a single wire connector may be multiple wire connector types; for example, wire connector A could be both type 2 720 and type 3 725.

FIG. 8 shows a module 800 that a may be used in systems and methods disclosed herein. Modules may built with wire connection portions (also known as wiring pins or points), which may be connectors, to connect wires from devices to the controller through the modules. Wire connection portions may be built with lever locks to lock a wire in place that will be connected to the module, and through the module wire connection portion, to a device. These wire connection portions may be connected by connectors, such as pogo pins, to the controller, which will pass information from the wire connected to the device, to the controller. In some embodiments, the wire connection portions comprise are tool-less lever lock connectors that securely connect almost any wire, from 10 to 26 gauge. The module 800 shows an embodiment with tool-less lever lock dry contacts 805, 810. With them, a level/lever 805 can be pulled down. A wire can then be inserted at 810. Pulling the level up will then pinch/secure the wire into the contact.

In this illustrative embodiment, wiring pins 810 have built-in line testing. When an installer pulls a wire and inserts it in the wire connection portion 810, software associated with the controller, the module, or a combination of the module and the controller will test and validate whether the pulled wire corresponds to the right wire in the model, and then indicate the correct correspondence in a visible way. In an illustrative embodiment, the module has LEDs positioned by the wires (or a selection of wires, or a single LED) that will indicate if the proper wire has been inserted, as shown at 815. In an embodiment, the LED flashes green if the proper wire has been pulled, or red if the wire is incorrect.

FIG. 9 shows an exemplary user interface with which devices can be added or modified by a controller or a system of controllers. The exemplary user interface shows the possible controller wiring connections 910 that devices can be hooked up to. In this embodiment, the display 900 shows a controller with eight modules 920 each of which have 6 wiring connections 910 where devices can be wired directly to the controller. Devices 905 can be moved into location on the controller display 900. The individual aspects of the wiring connections 915 are also displayed. In some embodiments, devices can be moved, the specifics of the devices can be modified, including chaining the nature of the wiring connections 915.

FIG. 10 shows some components that may be included in a computing environment used in the systems and methods described herein. The computing environment (which may be a controller or a controller system) may include at least one central processing unit 1005 and memory 1015, 1025. The processing unit may execute computer-executable instructions and may be a real or a virtual processor. There might also be a vector or co/processing unit 1010 that enables fast vector processing. In a multi-processing system, multiple processing units execute computer-executable instructions to increase processing power. The memory 1015, 1025 may be volatile memory (e.g., registers, cache, RAM), non-volatile memory (e.g., ROM, EEPROM, flash

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memory, etc.), or some combination of the two. For example, the memory can be volatile memory, e.g., static memory cells, as in FPGAs and some CPLDs; or non-volatile memory, e.g., FLASH memory, as in some CPLDs, or in any other appropriate type of memory cell. The memory stores software implementing described techniques and tools. The computer system may be distributed, with multiple processors and associated memory in different locations that communicate using wired or wireless network connections. These distributed computing nodes may run simultaneously run the same program using distributed computing techniques.

A computing environment may have additional features. For example, the computing environment may include storage **1020** which may also include memory **1025**, one or more input devices **1030**, one or more output devices **1035**, and one or more other communication devices **1040**. These may include touch screens, keyboards, game controllers, touchpads, LED screens, voice-operated input systems, printers, phone connections, FAX machines, etc. An interconnection mechanism such as a bus, controller, or network interconnects the components of the computing environment. Typically, operating system software stored in memory **1015**, **1025** provides an operating environment for other software executing in the computing environment, and coordinates activities of the components of the computing environment. The computer system **1000** may connect to other computer systems through network(s) **1050**, which may be wired, wireless, or both. Peripherals **1055**, such as external hard drives, modems, mice, keyboards, zip drives, scanners, 3-D printers, etc., may be included. Certain components may belong to more than one category.

The computing system **1000**, like other suitable systems, also includes one or more computer-readable storage media **1060**. Media **1060** may be of different physical types. The media **1060** may be volatile memory, non-volatile memory, fixed in place media, removable media, magnetic media, optical media, solid-state media, and/or of other types of physical durable storage media (as opposed to merely a propagated signal). In particular, a configured medium **1060** such as a portable (i.e., external) hard drive, CD, DVD, memory stick, or other removable non-volatile, non-transient memory medium may become functionally a technological part of the computer system when inserted or otherwise installed, making its content accessible for interaction with and use by central processing unit **1005**. The removable configured medium **1060** is an example of a computer-readable storage medium **1060**. Some other examples of computer-readable storage media **1060** include built-in RAM, ROM, hard disks, and other memory storage devices which are not readily removable by users **1045**. A computer-readable medium should not be considered a signal; neither should a computer-readable memory be considered a signal.

The medium **1060** is configured with instructions **1070** that are executable by a central processing unit **1005**; "executable" is used broadly to include, human readable source code, such as Java or C++, compiled code, and/or machine code. Executable code also includes code that runs using a distributed system, such as a series of controllers and controllers that distribute and run complex problems. The medium **1060** is also configured with data **1065** which is created, modified, referenced, and/or otherwise used for technical effect by execution of the instructions **1070**. The instructions **1070** and the data **1065** configure the memory or other storage medium **1060** in which they reside; when that memory or other computer readable storage

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medium is a functional part of a given computer system, the computer system may be configured by the instructions **1070** and data **1065**.

Although an embodiment may be described as being implemented as software instructions executed by one or more processors in a computing device (e.g., general purpose computer, cell phone, or controller), such description is not meant to exhaust all possible embodiments. One of skill will understand that the same or similar functionality can also often be implemented, in whole or in part, directly in hardware logic, to provide the same or similar technical effects. Alternatively, or in addition to software implementation, the technical functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without excluding other implementations, an embodiment may include hardware logic components such as Field-Programmable Gate Arrays (FPGAs), Application-Specific Integrated Circuits (ASICs), Application-Specific Standard Products (ASSPs), System-on-a-Chip components (SOCs), Complex Programmable Logic Devices (CPLDs), and similar components. Components of an embodiment may be grouped into interacting functional modules based on their inputs, outputs, and/or their technical effects, for example.

We claim:

1. A controller module comprising:

a frame;

a circuit board; disposed on the frame;

a connection to a controller;

a plurality of device connectors mounted on the frame, the plurality of device connectors operably able to connect a device to the controller module through a device wire; and

the circuit board executing a process which includes:

detecting that the device wire has been coupled to one of the plurality of device connectors;

sending information to the controller received from the device through the device wire;

receiving information from the controller indicating device validity; and

reporting the device validity;

line testing technology; and

wherein the line testing technology comprises at least one of: a ground connection, a 24 VAC power connection, a DAC with 0-10V DC analog voltage, a FET with a 1-wire pullup, a high range voltage divider, a low range voltage divider, a 4-20 mA current shunt, a 24 VAC current shunt, a DC offset injector, an electromechanical relay, a 120 VAC output, a current-measuring chip, a control for high current motors, or a feature that can switch 250 volt AC up to 10 amps.

2. The controller module of claim 1, wherein sending information to the controller comprises sending at least one of: voltage on the device wire; current on the device wire; protocol on the device wire, signal received from the device wire in response to a signal sent by the controller, or fault on the device wire.

3. The controller module of claim 1, further comprising operational ability to turn on the device associated with the device wire and operational ability to determine how much power is being drawn.

4. The controller module of claim 1, further comprising operational ability to determine voltage on the device wire when the device associated with that device wire is turned off.

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5. The controller module of claim 1, further comprising operational ability to change characteristics of the device wire.

6. The controller module of claim 5, wherein further comprising operational ability to change characteristics of the device wire comprises the controller module modifying hardware associated with the device wire.

7. The controller module of claim 6, further comprising operational ability to determine a voltage, a current, a signal, or a fault.

8. The controller module of claim 1, further comprising operational ability to report determined validity further comprises a light signaling when the information is valid.

9. A method performed by a controller module for line testing a device, the method comprising:

noticing a device has been operably attached to a wiring pin of a controller module;

sending a signal from the wiring pin to a controller attached to the controller module;

receiving a signal from the controller reporting on validity of the device with reference to the wiring pin;

reporting validity of the device using a reporting feature of the controller module, and

wherein the reporting feature comprises a light associated with the wiring pin.

10. The method of claim 9, wherein sending a signal from the wiring pin further comprises sending current flowing through the wiring pin.

11. The method of claim 10, further comprising changing a wiring feature of the wiring pin to match a wiring feature of the device.

12. The method of claim 11, wherein the wiring feature comprises a wiring fault, a power, a voltage, a current, or a protocol.

13. The method of claim 9, wherein reporting the validity using the reporting feature comprises turning on the light.

14. The method of claim 9, further comprising line testing.

15. The method of claim 14, wherein line testing comprises at least one of: testing for a ground connection, testing for a 24 VAC power connection, testing for a DAC with 0-10V DC analog voltage, testing for a FET with a 1-wire pullup, testing for a high range voltage divider, testing for a low range voltage divider, testing for a 4-20 mA current shunt, testing for a 24 VAC current shunt, testing for a DC

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offset injector, testing for an electromechanical relay, testing for a 120 VAC output, testing for a current-measuring chip, testing for a control for high current motors, or testing for a feature that can switch 250 volt AC up to 10 amps.

16. A non-transitory computer-readable storage medium configured with instructions for execution by a processor which upon execution performs a method of line testing a device, the method comprising:

instructions for noticing a device has been operably attached to a wiring pin of a controller module;

instructions for sending a signal from the wiring pin to a controller attached to the controller module;

instructions for receiving a signal from the controller reporting on validity of the device with reference to the wiring pin;

instructions for reporting validity of the device using a reporting feature of the controller module,

instructions for line testing, and

wherein line testing comprises at least one of: testing for

a ground connection, testing for a 24 VAC power

connection, testing for a DAC with 0-10V DC analog

voltage, testing for a FET with a 1-wire pullup, testing

for a high range voltage divider, testing for a low range

voltage divider, testing for a 4-20 mA current shunt,

testing for a 24 VAC current shunt, testing for a DC

offset injector, testing for an electromechanical relay,

testing for a 120 VAC output, testing for a current-

measuring chip, testing for a control for high current

motors, or testing for a feature that can switch 250 volt

AC up to 10 amps.

17. The non-transitory computer-readable storage medium of claim 16, further comprising instructions for changing characteristics of the wiring pin.

18. The non-transitory computer-readable storage medium of claim 16, further comprising instructions for modifying hardware associated with the wiring pin.

19. The non-transitory computer-readable storage medium of claim 16, further comprising instructions for modifying voltage expected for the wiring pin.

20. The non-transitory computer-readable storage medium of claim 16, further comprising wherein instructions for reporting validity comprises reporting that a light associated with the wiring pin has changed state.

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